

Week 12 Homework Assignment

Introduction to Functions

Total Points: 100

Name: _____

Class: _____

Important Instructions

- Work on your own, but you may ask family members for hints only if stuck
- Write your code in Python (*using Thonny or any online IDE*)
- For setting up your code files, kindly see [instructions.pdf](#)
- For written answers, use the spaces provided on this sheet
- Show your work for full credit!

1 Part 1: Function Foundations (15 points)

Time estimate: 15 minutes

1.1 Exercise 1.1: Function Anatomy (8 points)

Label each part of this function definition by writing the correct term in the blank:

```
def calculate_total(price, quantity):  
    result = price * quantity  
    return result
```

Code Element	What is it called?
def	_____
calculate_total	_____
price, quantity	_____
: (colon after parentheses)	_____
result = price * quantity	_____
return result	_____

1.2 Exercise 1.2: Multiple Choice (7 points)

Circle the correct answer for each question:

1. What happens if you forget the parentheses when calling a function?

- A. Python shows a syntax error
- B. The function runs normally
- C. Python refers to the function object but doesn't execute it
- D. The program crashes immediately

2. What does a function return if it has no `return` statement?

- A. 0
- B. An empty string
- C. None

D. An error

3. Which is the correct way to define a function?

A. `function greet():`

B. `def greet():`

C. `define greet():`

D. `func greet():`

2 Part 2: Code Tracing with Scope (20 points)

Time estimate: 20 minutes

2.1 Exercise 2.1: Trace the Execution (10 points)

Trace through this code step by step. Fill in the memory diagram and predict the output.

```
def add_bonus(score, bonus):  
    total = score + bonus  
    return total  
  
points = 50  
extra = 15  
final = add_bonus(points, extra)  
print(f"Final score: {final}")
```

Memory Diagram: Complete the diagram showing the state just BEFORE the function returns:

Global Frame

points = _____
extra = _____
final = _____

add_bonus Frame

score = _____
bonus = _____
total = _____

Predicted Output: _____

2.2 Exercise 2.2: Scope Detective (10 points)

This code has a scope-related bug. Study it carefully, then answer the questions.

```
def calculate_discount(price):  
    discount = price * 0.10  
    final_price = price - discount  
    return final_price  
  
original = 500  
result = calculate_discount(original)  
print(f"You saved: {discount} rupees") # Error occurs here!
```

a) What error will Python show? _____

b) Why does this error occur? (Explain in terms of scope)

c) Write the corrected version of this code:

Corrected code:

3 Part 3: Code Completion (18 points)

Time estimate: 20 minutes

3.1 Exercise 3.1: Complete the Function (9 points)

Fill in the blanks to create a function that calculates the area of a rectangle:

```
# Function to calculate rectangle area
----- rectangle_area(length, width):
    area = ----- * -----
    ----- area

# Test your function
result = rectangle_area(10, 5)
print(f"The area is: {result} square units")
```

Save this code as week12_ex3_1.py

3.2 Exercise 3.2: Build a Greeting Function (9 points)

Complete this function that creates a personalized Pakistani greeting:

```
# Function to create a greeting message
def create_greeting(_____, _____):
    # Combine name and city into a greeting
    message = f"Assalam-o-Alaikum, {_____}! Welcome from {_____}!"
    _____ message

# Test with different values
greeting1 = create_greeting("Ahmed", "Lahore")
greeting2 = create_greeting("Fatima", "Karachi")

print(greeting1)
print(greeting2)
```

Save this code as week12_ex3_2.py

4 Part 4: Debugging Functions (15 points)

Time estimate: 15 minutes

Each code snippet has errors. Find and fix ALL errors:

4.1 Snippet 1: Missing Pieces (5 points)

```
# This code has 3 errors - fix them all!
def double_number(n)
    result = n * 2
    return result

answer = double_number 10
print(answer)
```

Fixed version:

```
# Write the corrected code here:
```

4.2 Snippet 2: Return vs Print Confusion (5 points)

```
# This code has 2 errors - fix them all!
def calculate_average(a, b, c):
    avg = (a + b + c) / 3
    print(avg)

result = calculate_average(80, 90, 85)
final = result + 5 # We want to add bonus points
print(f"With bonus: {final}")
```

Fixed version:

```
# Write the corrected code here:
```

4.3 Snippet 3: Parameter Mismatch (5 points)

```
# This code has 2 errors - fix them all!
def greet_student(name, grade, school):
    print(f"{name} is in grade {grade} at {school}")

greet_student("Ali", 7) # Missing an argument!
greet_student("Sara", "MAKTAB", 8) # Wrong order!
```

Fixed version:

```
# Write the corrected code here:
```

Save all corrected code as `week12-ex4-fixed.py`

5 Part 5: Full Program Development (22 points)

Time estimate: 25 minutes

5.1 Exercise 5.1: Build a Grade Calculator (22 points)

Create a program called `grade_calculator.py` that uses functions to calculate and display student grades.

Program Requirements

Your program should:

- Define a function `calculate_percentage(obtained, total)` that returns the percentage
- Define a function `get_grade(percentage)` that returns a letter grade (A, B, C, D, F)
- Define a function `display_result(name, percentage, grade)` that prints a formatted result
- Use all three functions together in the main program
- Handle at least 2 different students

Grade Scale:

- A: 90-100%
- B: 80-89%
- C: 70-79%
- D: 60-69%

- F: Below 60%

Example Run:

```
===== Grade Calculator =====
Student: Ahmed
Marks Obtained: 85 out of 100
Percentage: 85.0%
Grade: B
-----
```

```
Student: Fatima
Marks Obtained: 92 out of 100
Percentage: 92.0%
Grade: A
```

Starter Structure:

```
# grade_calculator.py
# Name: -----
# Date: -----

# Function to calculate percentage
def calculate_percentage(obtained, total):
    -----
    -----

# Function to determine letter grade
def get_grade(percentage):
    -----
    -----
    -----
    -----
    -----

# Function to display the result
def display_result(name, percentage, grade):
    -----
    -----
    -----

# Main program
print("===== Grade Calculator =====")

# Student 1
name1 = "Ahmed"
marks1 = 85
total1 = 100
pct1 = -----
grade1 = -----
-----

print("-----")

# Student 2
name2 = "Fatima"
marks2 = 92
total2 = 100
-----
-----
-----
```

Testing Checklist:

- ☐ Program runs without errors
- ☐ `calculate_percentage` returns correct values
- ☐ `get_grade` returns correct letter grades for all ranges
- ☐ `display_result` prints formatted output
- ☐ All three functions work together correctly
- ☐ Tested with at least 2 students

Save this program as `grade_calculator.py`

6 Part 6: Reflection Questions (5 points)

Time estimate: 10 minutes

Answer these questions in complete sentences:

1. **Why are functions useful?** Think about a long program without functions versus one with functions.

2. **Explain the difference between `return` and `print()` in your own words.** When would you use each one?

3. **What is scope?** Why can't you use a variable from inside a function in your main program?

7 Part 7: Creative Challenge (5 points)

Time estimate: 15 minutes

Choose ONE of the following mini-projects:

7.1 Option A: Cricket Score Calculator

Create functions to calculate cricket statistics:

```
# Define these functions:
# - calculate_run_rate(runs, overs) -> returns run rate
# - calculate_required_rate(target, runs, overs_left) -> returns required
  rate
# - display_match_status(team, runs, overs, target) -> prints formatted
  status

# Test with sample match data
```

7.2 Option B: Simple Unit Converter

Create functions for common Pakistani conversions:

```
# Define these functions:
# - km_to_miles(km) -> returns miles
# - kg_to_pounds(kg) -> returns pounds
# - celsius_to_fahrenheit(c) -> returns fahrenheit
# - display_conversion(value, from_unit, result, to_unit) -> prints nicely

# Test each conversion function
```

Save your chosen project as `week12_creative.py`

8 Debugging Discoveries (5 bonus points)

Growth Mindset: Errors are learning opportunities!

Did you encounter any interesting errors while completing this homework? Share one error and how you fixed it:

My Debugging Story

The Error I Got: _____

What I Tried First: _____

How I Fixed It: _____

What I Learned: _____

No errors? Share a tricky part you figured out instead!

9 Bonus Section: Extra Challenges (Optional - 5 extra points)

Only attempt if you've finished everything else!

9.1 Challenge: Function Composition (5 points)

Create a chain of functions where one function calls another:

```
# Create these functions:
# 1. square(n) - returns n squared
# 2. add_ten(n) - returns n + 10
# 3. process_number(n) - squares the number, then adds ten to the result

# Example: process_number(5) should return 35 (5^2 = 25, 25 + 10 = 35)

# Your code here:


# Test it:
print(process_number(5))    # Should print 35
print(process_number(3))    # Should print 19
```

Save as week12_bonus.py

Submission Checklist

Before submitting, make sure you have:

- ☐ Completed all required sections (Parts 1-7)
- ☐ Saved each code exercise in its own file:
 - `week12_ex3_1.py` (rectangle area)
 - `week12_ex3_2.py` (greeting function)
 - `week12_ex4_fixed.py` (debugged code)
 - `grade_calculator.py` (main program)
 - `week12_creative.py` (creative challenge)
- ☐ Created a folder named `week12_hw_yourname` with all files
- ☐ Tested each code file individually to ensure it runs without errors
- ☐ Verified all functions have proper `return` statements where needed
- ☐ Checked that function calls include parentheses
- ☐ Written your name at the top of this paper
- ☐ Answered all reflection questions
- ☐ Attempted bonus section (optional)

Week 12 Concepts Check:

Which function concept was most challenging? _____

What would you like more practice with? _____

Time Tracking:

How long did this homework take you? _____ hours

Parent/Guardian Signature: _____

(Confirming student completed work independently)

Great job learning about functions!

Next week: Functions calling functions and building complete multi-function programs